DEFENSE & SYSTEMS ECONOMICS

# Turkey's Simplicity Versus Integration

Why Two Design Philosophies for Military Power Are Diverging, Not Converging

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**IDEA IN BRIEF**

**THE OBSERVATION**
In defense technology, the cost of the most sophisticated platforms and the cost of the simplest effective ones are diverging, not converging. The F-35 programme's lifecycle cost has risen to $1.7 trillion. The weapon doing equivalent tactical work in Ukraine costs $500. The gap has widened by four orders of magnitude in a single decade.

**THE EXPLANATION**
The divergence is structural, not accidental. Exquisite platforms demand coordination architectures—data links, identification systems, doctrinal harmonization, logistics chains—whose costs scale non-linearly with complexity. Simple platforms are designed to minimize coordination requirements. One paradigm adds architectural layers and pays the scaling penalty; the other strips them away and captures the savings.

**THE IMPLICATION**
If the cost curves continue to diverge, the scenarios in which expensive integrated platforms justify their cost will narrow with each passing year. Countries that bet on the integration paradigm—including the one whose drones did more than any other to prove the simplicity paradigm—face a strategic dilemma that spending alone cannot resolve.


In most industries, we expect cost convergence. A cheaper alternative enters the market and improves until it matches the incumbent, or the incumbent learns from the challenger and brings its costs down. Competition narrows the gap. This is one of the most basic assumptions in economics, and for most goods and services it holds.

In defense technology, the opposite is happening.

The Bayraktar TB2 drone entered Turkish military service in 2014 at a unit cost of roughly five million dollars. A decade later, the weapon doing equivalent tactical work on the front lines of Ukraine—a first-person-view kamikaze drone—costs between five hundred and a thousand dollars. Meanwhile, the F-35 Lightning II has seen its total programme lifecycle cost rise to an estimated $1.7 trillion.

The gap between the most expensive approach to airpower and the cheapest effective one has not narrowed. It has widened by roughly four orders of magnitude in ten years. And the rate of divergence is accelerating.

## TWO PARADIGMS

The pattern has a longer history than the drone age. It traces back at least to the Second World War, and it reflects a deliberate design philosophy rather than an accident of cost engineering.

The Soviet T-34 tank was not the best tank of the war. The German Tiger I was heavier, better armed, and more precisely engineered, but the T-34 was designed from the outset for mass production: wide tracks that handled mud without engineering support, a diesel engine that ran on fuel available everywhere, welded armor that could be assembled by semi-skilled workers in relocated factories. The Soviets accepted a cramped turret, poor gun sights, and higher crew casualties because those trade-offs bought a tank that could be produced at roughly five times the rate of the Tiger. This was not a compromise; it was intentionally engineered to build a platform so simple that the coordination architecture around it—the logistics tail, the maintenance

infrastructure, the training pipeline—approaches zero.

A senior figure at one of Turkey's largest weapons manufacturers has described the company's approach as explicitly modeled on this tradition: build for production volume, accept lower individual performance, and design out the need for complex support infrastructure. The TB2 embodies this.

This paper terms this the **simplicity paradigm**. It runs from the T-34 through the AK-47 to the TB2, and now to the FPV drone. Each generation strips away more of the coordination burden, and the cost follows.

The opposing approach—call it the **integration paradigm**—runs from the Tiger tank through the F-15, the Patriot missile system, and the F-35. Each platform demands an increasingly elaborate architecture to function: shared data links, identification-friend-or-foe systems, doctrinal harmonization, dedicated logistics chains, and institutional machinery to keep heterogeneous systems behaving as a coherent whole.

These two paradigms have coexisted for decades. What is new is that their cost curves are moving in opposite directions, and the divergence is structural.

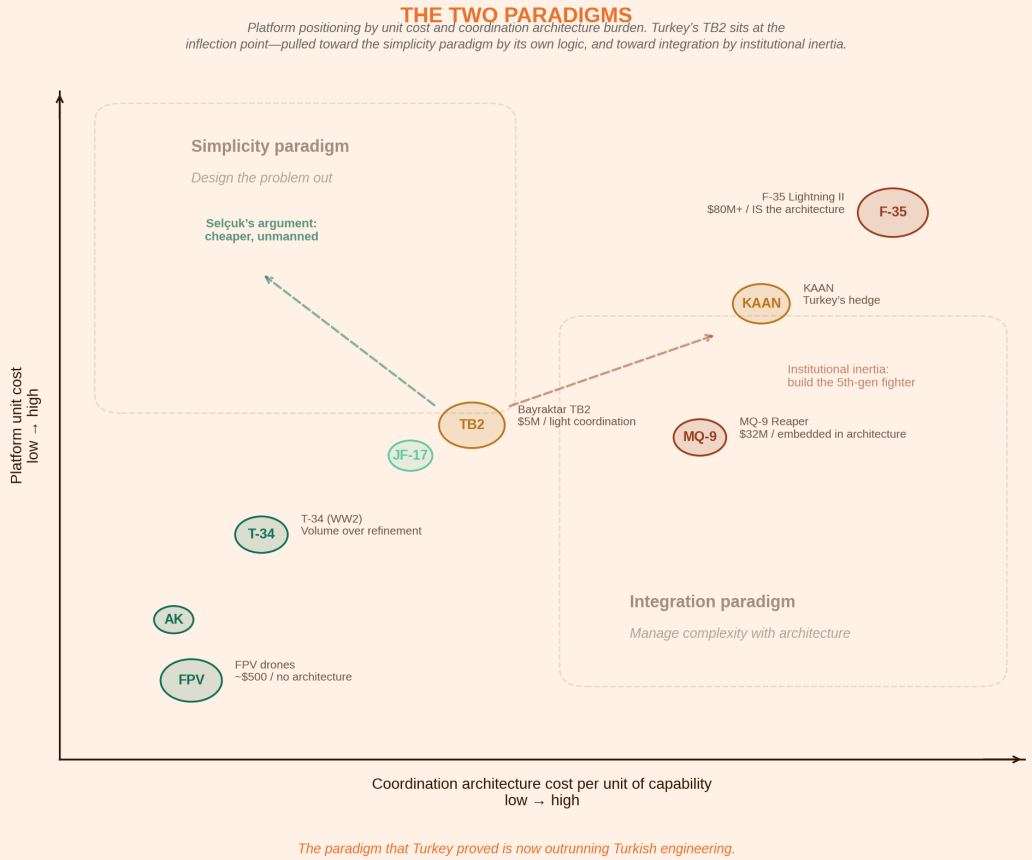



*Figure 1. Platform positioning by unit cost and coordination architecture burden. Turkey's TB2 sits at the inflection point—pulled toward cheaper unmanned platforms by its own logic, and toward the KAAN by institutional inertia.*

## WHY DIVERGENCE, NOT CONVERGENCE

The *integration paradigm's* costs rise because coordination architecture costs scale non-linearly with complexity. When a military force must coordinate capabilities across air, land, sea, space, cyber, and the electromagnetic spectrum, the number of interfaces grows combinatorially. With two domains, there is one interface. With six, there are fifteen—each requiring its own data protocols, timing synchronization, identification procedures, and rules of engagement.

The real cost is in the multi-way interactions: an electronic-warfare operation that jams enemy radar

also degrades friendly identification systems; a cyber operation that disrupts adversary command networks may affect the timing of a coordinated air-ground strike. These costs compound with every platform, every domain, and every coalition partner.

The F-35 is the purest expression of this logic. It is not just a fighter aircraft; it is a data node embedded in an architecture of shared intelligence, logistics, and multinational doctrine. Its cost includes that architecture's maintenance. And the architecture's cost has grown faster than the platform's capability.

The *simplicity paradigm's* costs fall for the mirror-image reason; that each generation minimizes the coordination architecture it requires. The FPV drone needs a camera, a motor, a battery, a 3D-printed frame, and a warhead. The coordination burden approaches zero, and the unit cost follows it down.

## THE COST-EXCHANGE SPIRAL

The divergence has a direct operational consequence: the cost-exchange ratio between attacker and defender is not stable. It is worsening over time, and in a single direction.

In Libya in 2020, each Turkish drone that destroyed a Pantsir air-defense system generated a ratio of roughly three to one. In Nagorno-Karabakh, eighty to one. By 2024 in Ukraine, defending against a two-hundred-dollar FPV drone with a three-million-dollar Patriot interceptor produced a ratio of fifteen thousand to one.

These create a trend line whereby the ratio is accelerating because the cost of the integration paradigm's defense is rising, while the cost of the simplicity paradigm's attack is falling.

The production arithmetic compounds the problem. Global Patriot interceptor production runs to roughly 650 missiles per year; FPV drone production in Ukraine alone runs to millions. Even perfect interception accuracy cannot sustain a defense whose unit cost exceeds the attacker's by four orders of magnitude at volumes the attacker can outproduce by three.

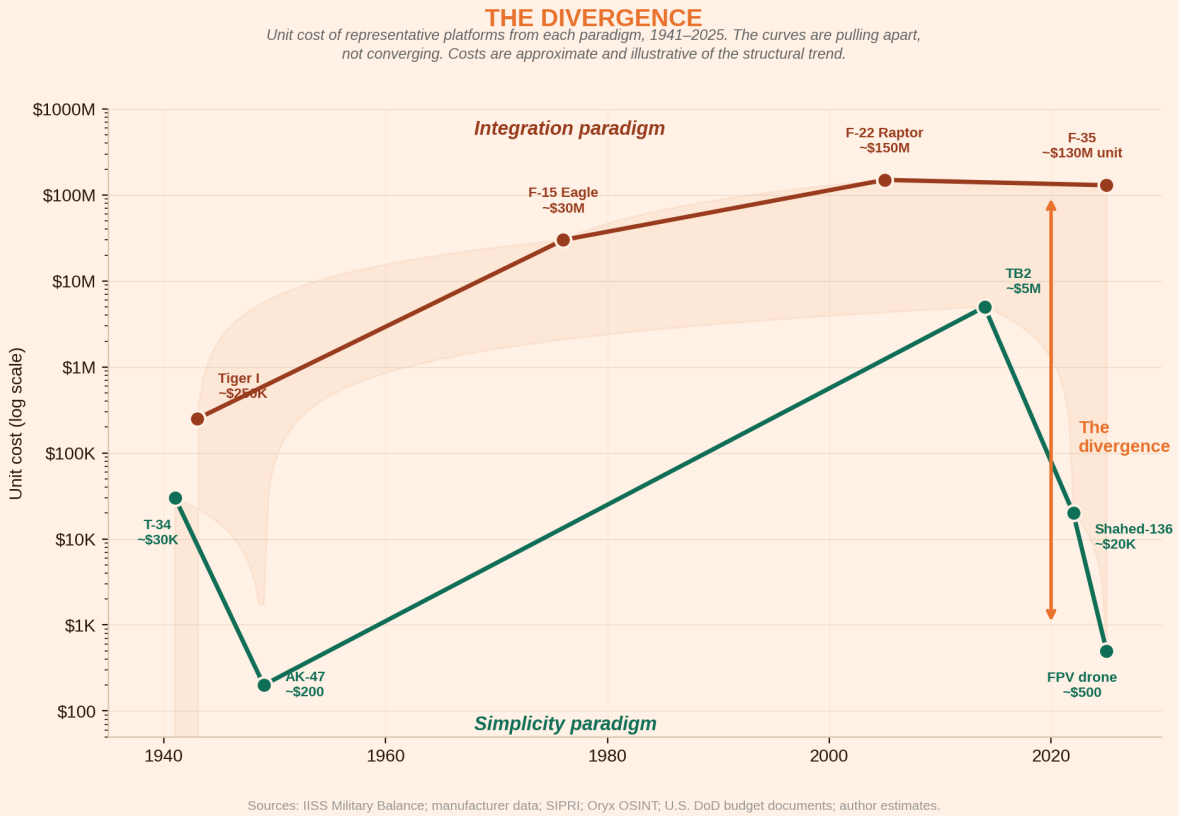



*Figure 2. Unit cost of representative platforms from each paradigm, 1941–2025. The integration paradigm's cost rises; the simplicity paradigm's falls. Sources: IISS Military Balance; manufacturer data; SIPRI; Oryx; U.S. DoD budget documents; author estimates.*

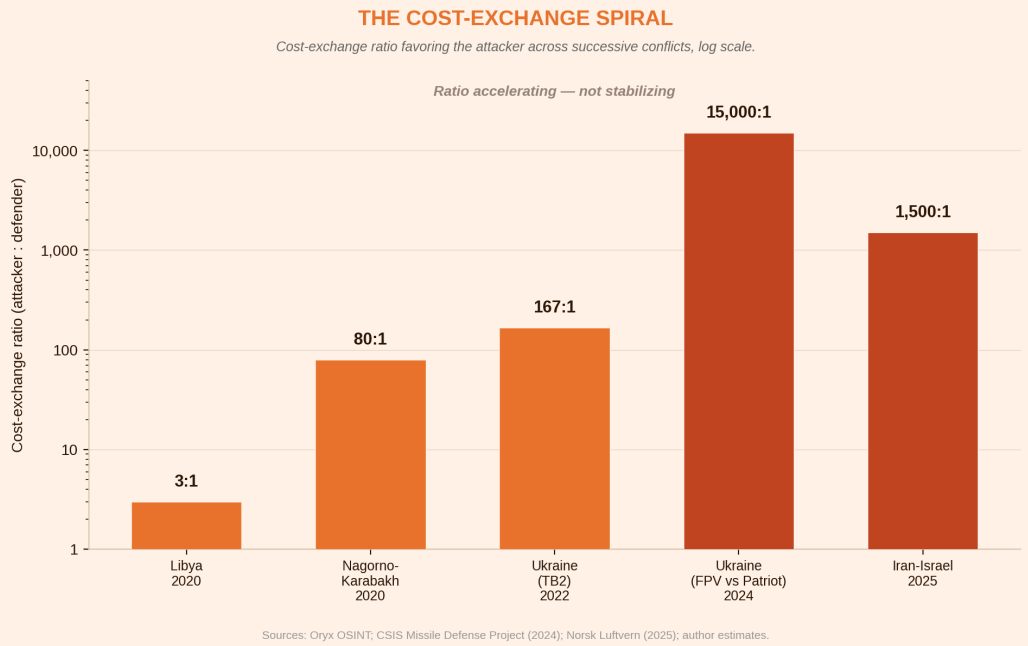



*Figure 3. Cost-exchange ratio favoring the attacker across successive conflicts. The ratio is accelerating by orders of magnitude. Sources: Oryx OSINT; CSIS Missile Defense Project (2024); Norsk Luftvern (2025); author estimates.*

*The integration paradigm's costs rise because each generation adds complexity. The simplicity paradigm's costs fall because each generation strips it away. The gap is structural, and it is accelerating.*

## BEYOND DEFENSE

The divergence is not confined to weapons systems. A structurally identical pattern is playing out in the semiconductor industry, where the cost of leading-node fabrication and the cost of trailing-node production are pulling apart on a similar trajectory.

A leading-edge 2nm fabrication facility now costs upward of twenty billion dollars to build. The coordination architecture—from extreme ultraviolet lithography sourced through a single Dutch supplier to cleanroom environments operating at tolerances measured in atoms—is enormous. This is the semiconductor equivalent of the F-35.

Meanwhile, China is building an industrial strategy around trailing-node chips—the 28nm and 14nm processes that power automobiles, industrial equipment, and the vast majority of AI inference workloads. The cost per transistor at trailing nodes continues to fall. The cost per transistor at leading nodes has begun to rise.

The parallel is structural. In both defense and semiconductors, the exquisite solution's cost is rising because the coordination architecture it demands is growing more complex. The good-enough solution's cost is falling because its architecture requirements are shrinking.

## TURKEY'S INSTITUTIONAL DIVIDE

No country illustrates the tension between these paradigms more clearly than Turkey—not because Turkey is caught between them, but because two of its institutions have placed opposing bets.

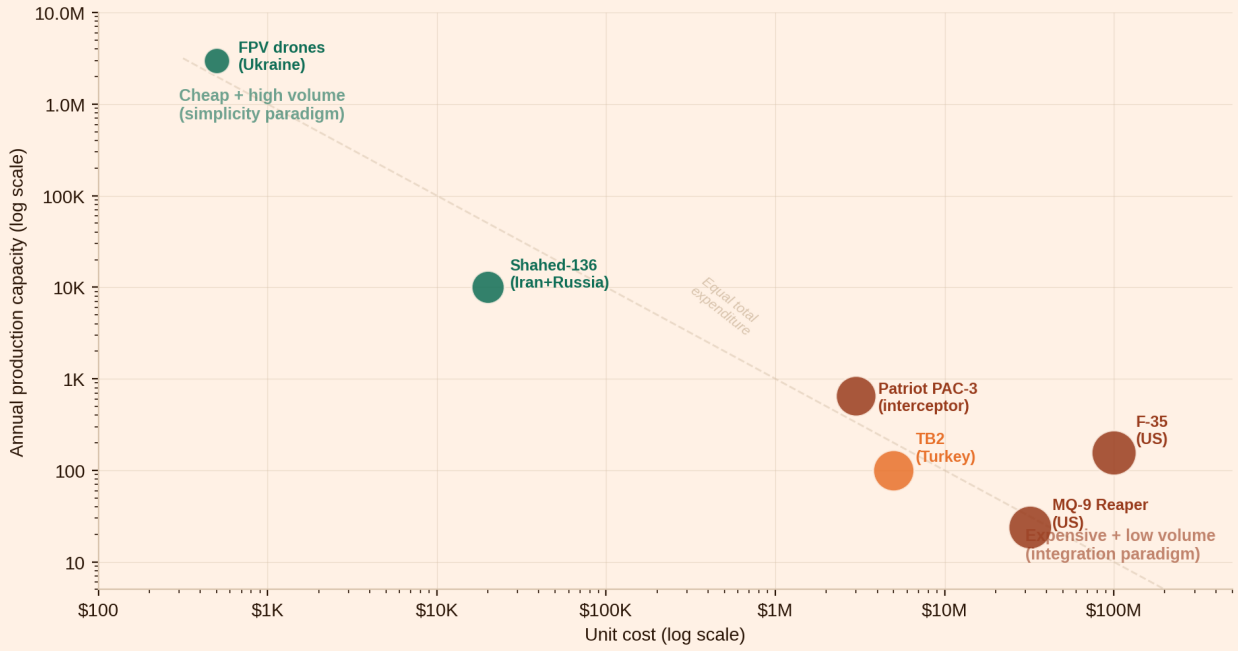
Baykar, the company that built the TB2, is firmly in the simplicity paradigm and doubling down. The TB2 demonstrated that a five-million-dollar drone could destroy a fourteen-million-dollar air-defense system. When the paradigm moved past the TB2 toward cheaper platforms—millions of FPV drones at a fraction of its cost—Baykar adapted, developing upgraded variants and carrier-capable systems while its founder's son, Selçuk Bayraktar, publicly argued for a national strategy centered on low-cost unmanned platforms equipped with artificial intelligence. Baykar's position is not ambivalent. It is a deliberate, stated commitment to the paradigm the company proved.

Turkish Aerospace Industries, the state-owned enterprise responsible for the KAAN fifth-generation fighter, is pursuing the opposite strategy. The KAAN is a Turkish airframe with Turkish avionics and Turkish weapons—an astonishing achievement of indigenous capability—but it is also an integration paradigm platform that demands the kind of coordination architecture whose costs the divergence is driving upward. Its American engine remains blocked by Congress, and its strategic rationale rests on the assumption that manned, stealthy, highly integrated aircraft will remain necessary despite the evidence from Turkey's own wars.

These are not two sides of the same hedge; they are two institutions within the same state making structurally incompatible bets about the future of military power. Baykar is betting that the simplicity paradigm's cost curve will continue to fall and that the value lies in volume, attritability, and software autonomy. TAI is betting that the integration paradigm still has a future worth investing billions in. The divergence suggests that one of these bets is right and the other is a misallocation of national capital—and the evidence, including evidence generated by Turkey's own weapons in other countries' wars, increasingly favors Baykar's position.

**COST VERSUS VOLUME**

*Unit cost vs. annual production capacity for representative platforms. The simplicity paradigm occupies the upper left; the integration paradigm occupies the lower right. Both axes log scale.*




Sources: manufacturer data; Defense Express (2026); CSIS; U.S. DoD budget documents; author estimates.

*Figure 4. Unit cost vs. annual production capacity. The simplicity paradigm occupies the upper left; the integration paradigm the lower right. The TB2 sits in the middle. Sources: manufacturer data; Defense Express (2026); CSIS; U.S. DoD budget documents; author estimates.*

## THE STRATEGIC CALCULUS

The divergence does not argue that expensive platforms are useless, as a force that can coordinate across domains retains enormous advantages. But the question is whether those advantages can keep pace with the widening cost differential. The integration paradigm does not need to become *ineffective* for it to become *unaffordable*.

Baykar designed the weapon that proved the simplicity paradigm could work. The question now is not whether Baykar's bet is correct—the evidence increasingly suggests it is—but whether the institutions allocating national defense capital in Turkey, the United States, and across NATO will follow the cost curve or continue investing in platforms whose coordination architecture costs are rising faster than the capability they deliver.

## IMPLICATIONS FOR FORCE DESIGN

**1. Procurement must measure architecture cost, not just platform performance.** Current systems evaluate platforms against domain-specific requirements without measuring the platform's demand on the coordination architecture that makes it operationally useful. If the divergence is structural, procurement that ignores architecture cost will systematically over-invest in the paradigm whose costs are rising.

**2. Force structure should be tiered, not uniform.** The implication is a two-tier force: an exquisite core reserved for missions where multi-domain coordination is genuinely irreplaceable—strategic deterrence, deep strike, contested space—surrounded by a mass layer of cheap, attritable, loosely coordinated systems. The cost-exchange arithmetic is indifferent to institutional preference.

**3. The binding constraint is production capacity, not unit performance.** Patriot interceptor production at 650 per year against millions of attack drones is not a technology gap; it is an industrial architecture gap. The capacity to produce adequate systems at scale matters more than the ability to produce superior systems in limited numbers.

**4. National capital allocation should follow the cost curve, not institutional prestige.** Turkey's experience is instructive: Baykar's simplicity-paradigm platforms have generated demonstrated combat results across multiple theatres, while the state-owned KAAN programme consumes billions pursuing the integration paradigm on which the divergence is closing. The capital allocated to the KAAN by Turkish Aerospace Industries could fund an ecosystem of cheap, AI-enabled, attritable unmanned platforms—the direction Baykar is already moving. More broadly, mid-tier defense producers face a choice between competing at the frontier, where costs are escalating beyond their fiscal reach, and deepening their advantage where their cost structure is strongest. The divergence favors the latter.

**5. The next strategic advantage may be architectural, not manufacturing.** The simplicity paradigm's trajectory favors whoever can produce the most disposable platforms at the lowest cost—a competition in which Chinese manufacturing starts with a structural advantage. But platforms without coordination still need software autonomy, swarm logic, targeting algorithms, and integration standards. The country that designs the coordination layer for cheap, attritable systems may hold the defining advantage of the next era. This is the defense equivalent of designing the chip architecture rather than fabricating the chip.

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*This article draws on frameworks developed in two working papers by Sinéad O'Sullivan: **Institutions as Coordination Architectures** and **Market Formation as a Systems Engineering Problem**. Copies available at **s@sinead.co**.*